

Anharmonic phonon NEGF: the 3-ph bubble in production

SCBA self-energy (cubic FC $\Phi_{\mu\nu\nu'}$)

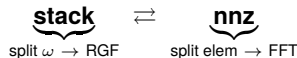
$$\Sigma_{\mu\mu'}^{\lessgtr}(\omega) = \frac{i\hbar}{2} \sum_{\nu_1\nu_2\nu_3\nu_4} \Phi_{\mu\nu_1\nu_2} \Pi_{\nu_1\nu_4,\nu_2\nu_3}^{\lessgtr}(\omega) \Phi_{\mu'\nu_3\nu_4}$$

$$\Pi^{\lessgtr}(\omega) = \mathcal{F}^{-1} \left[g^{\lessgtr}(\tau) g^{\lessgtr}(\tau) \right], \quad g^{\lessgtr} = \mathcal{F} \left[G^{\lessgtr} \right]$$

ring $\Phi \cdot [g g] \cdot \Phi$: 3 GEMMs, transpose-free

Production (src/quatrex, distributed)

Two layouts, swapped by all-to-all (dtranspose):



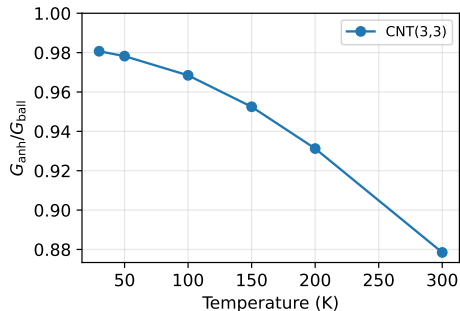
3 MPI axes $P = P_{\omega} \times P_{\text{blk}} \times P_{\text{ql}}$

SCBA iteration: **3 redistributions**

$$\text{RGF}_{\text{st}} \xrightarrow{1} \text{FFT}_{\text{nnz}} \xrightarrow{2} (\Phi \Pi \Phi)_{\text{st}} \xrightarrow{3} \text{IFFT}_{\text{nnz}} \rightarrow \text{RGF}$$

M. Luisier, PRB **86**, 245407 (2012) — SCBA 3-ph NEGF. Σ^R , $G^{\lessgtr} = G^R \Sigma^{\lessgtr} G^A$ built downstream.

CNT: channel freezing and length ladder (production)



- CNT(3,3): $G_{\text{anh}}/G_{\text{ball}}$ falls **0.981** $\rightarrow$ **0.878** (30 $\rightarrow$ 300 K) — ballistic limit recovered as channels freeze out
- length ladder ($L=2/3/4$ cells, 300 K): ratio 0.878 / 0.895 / 0.855 — onset of the resistive regime; $L \geq 4$ only reachable with the distributed solver
- $G_{\text{ball}}(300 \text{ K}) = 5.1 \times 10^8 \text{ W m}^{-2} \text{ K}^{-1}$, conservation $\sim 9\text{e-}05$
- CNT(8,0): 0.775 (L1, 300 K) (dense)

SiNW: soft modes need the static diagrams (loop + tadpole)

$$\Sigma_{\mu\mu'}^{\text{tad}} = \Phi_{\mu\mu'\nu} \langle u_\nu \rangle, \quad \langle u \rangle = -D^+ \left[\frac{1}{2} \Phi_{\nu\kappa\lambda} \langle u_\kappa u_\lambda \rangle \right]$$

$$\Sigma_{\mu\mu'}^{\text{loop}} = \frac{1}{2} \Phi_{\mu\mu'\kappa\lambda}^{(4)} \langle u_\kappa u_\lambda \rangle, \quad \langle uu \rangle = \int \frac{d\omega}{2\pi} iG^<$$

- d5a hosts an ultra-soft 0.0075 THz twist mode: tadpole alone *diverges*; **loop is the dominant correction** and stabilises the SCBA ($\|\Sigma_T\|$ reduced $4.6\times$)
- d5a: ratio 0.942 (30 K, $\lambda=0.3$) (dense)
- d11a (no soft mode): nearly ballistic, 0.997 ($\lambda=0.3$) (dense); multi-cell anharmonic now running (beyond dense memory wall)
- diagonal- G approximation fails $5\text{--}30\times$: off-diagonal coherence essential

fig/static_se_study.pdf

Si thin film: coupled- $q_{\perp}$ scattering vs. Guo 2020

$$\Sigma^{\lessgtr}(\mathbf{q}_{\perp}, \omega) = \sum_{\mathbf{q}'_{\perp}} \tilde{\Phi}(\mathbf{q}'_{\perp}, \mathbf{q}_{\perp} - \mathbf{q}'_{\perp}) [G^{\lessgtr}(\mathbf{q}'_{\perp}) G^{\lessgtr}(\mathbf{q}_{\perp} - \mathbf{q}'_{\perp})] \tilde{\Phi}^{\dagger}$$

- $\mathbf{q}'_{\perp}$ -convolution couples all transverse momenta (ballistic: independent $\mathbf{q}_{\perp}$) — dedicated $\mathbf{q}_{\perp}$ -MPI axis
- ballistic matches Guo's scale (~ 950 vs. $\sim 1030 \text{ MW m}^{-2} \text{ K}^{-1}$)
- anharmonic reduction at 1.5 nm: $\sim 45\%$ **vs. their** $\sim 10\%$ — open; prefactor and FC3 range ruled out
- reduction grows with thickness (correct diffusive trend)

fig/si_film_vs_guo.pdf

Y. Guo, M. Bescond *et al.*, PRB **102**, 195412 (2020): $\sim 20\%$ reduction at 10 nm/300 K, weaker at small thickness.

SrTiO₃ and what is running now

fig/srtio3_rattle_renorm.pdf

Production rerun status (auto-generated 2026-06-10 17:08)

system	pts	ratio	state
CNT(3,3)	9	0.878	done
CNT(8,0)	–	0.775	running
SiNW d5a	–	0.942	running
SiNW d11a	–	0.997	running
SrTiO ₃	–	–	running
Si film	–	0.52–0.38	running

Next

- film + SrTiO₃ sweeps land tonight
- solver audit done: one normalisation bug found & fixed (KK term; unused by production), coupled- $q_{\perp}$ fold being verified, film path parallelisation in progress